NAME:ALAN PIOSAM A
REG NO:411625104004
DEPARTMENT:CSE
SUBJECT:PYTHON
BATCH:03

STUDENT RECORD BY USING CSV FILE

AIM

To create a Python program using CSV file handling to store student records, display them, and search for a specific student by ID.

PROCEDURE

1. Import the csv module in Python.
2. Create a file named students.csv.
3. Write column headers (ID, Name, Age, Marks) into the file.
4. Insert student records into the CSV file.
5. Open the file in read mode and display all records.
6. Define a search ID to find a specific student.
7. Traverse the file and compare each record's ID.
8. Display the student details if found, otherwise print "Student not found".

CODE:

```
import csv

file_name = "students.csv"

# Create file and write header
with open(file_name, "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerow(["ID", "Name", "Age", "Marks"])

# Add student records (NEW NAMES)
with open(file_name, "a", newline="") as file:
    writer = csv.writer(file)
    writer.writerow([101, "Rahul", 20, 86])
    writer.writerow([102, "Sneha", 19, 91])
    writer.writerow([103, "Arjun", 21, 79])
```

```
print("Student Records")
print("---------------------")

# Read and display records
with open(file_name, "r") as file:
    reader = csv.reader(file)
    for row in reader:
        print(row)

# Search for a student
search_id = "102"
found = False

with open(file_name, "r") as file:
    reader = csv.reader(file)
    for row in reader:
        if row[0] == search_id:
            print("\nStudent Found:", row)
            found = True

if not found:
    print("Student not found")
```

OUTPUT:

Output

```
Student Records
---------------------
['ID', 'Name', 'Age', 'Marks']
['101', 'Rahul', '20', '86']
['102', 'Sneha', '19', '91']
['103', 'Arjun', '21', '79']

Student Found: ['102', 'Sneha', '19', '91']
```

RESULT

The program successfully created a CSV file, stored student data, and displayed all records. It also correctly searched and displayed the student with the given ID.

CONCLUSION

Thus, the use of CSV file handling in Python helps manage and retrieve student data efficiently.